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RESEARCH ARTICLE

Distributionally Robust and Learning-Based Vehicle Routing for Disaster Blood Logistics

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ABSTRACT In humanitarian disasters, delivering blood and blood products rapidly and reliably is critical but logistically complex. This paper proposes a hybrid framework that integrates Distributionally Robust Optimization (DRO) for offline anticipative planning with Contextual Bandits for online adaptive vehicle dispatch. The system coordinates a heterogeneous fleet of ambulances, drones, and robots to navigate uncertain demand, infrastructure degradation, and time-sensitive deliveries. We formulate a two-level approach: a robust vehicle routing problem accounting for moment-based uncertainty in demand, and an online dispatch policy that learns context-aware allocation decisions in real time. Extensive simulations evaluate the proposed method against DRO-only, CB-only, and stochastic routing baselines. Results show that the hybrid approach outperforms all others in robustness, service rate, and delay, while maintaining reasonable computational efficiency.

INDEX TERMS Blood supply chain, disaster, robust optimization, bandit learning, vehicle routing problem.

I. INTRODUCTION

Disaster scenarios pose extreme challenges for blood logistics, where timely delivery of blood products can mean the difference between life and death. In the aftermath of events such as earthquakes or hurricanes, local infrastructure is often severely damaged, hindering the distribution of blood to field hospitals and trauma centers when demand surges acutely [1], [2]. For example, analyses of past disasters have shown that while blood supplies might be adequate, the disruption of transportation networks becomes the critical bottleneck in meeting urgent transfusion needs [3]. Blood is a highly perishable medical commodity –red blood cells typically expire within 42 days and platelets within only 5 days – and cannot be stockpiled easily in unstable field conditions [4], [5]. By 2020, over 75% of Rwanda's emergency blood shipments were conducted via drone [6]. Thus, effective blood logistics in disasters require rapid, flexible, and reliable delivery under uncertain and evolving conditions [7].

Recent pilot programs have demonstrated the potential of innovative transport modes like drones to bypass damaged roads and dramatically cut delivery times for emergency blood shipments [8], [9]. For instance, integrating medical delivery drones in urban emergency networks has been shown to reduce transport times from over 90 minutes to 20 minutes during peak congestion [10]. These developments underscore a compelling motivation to explore multi-modal vehicle routing – coordinating ambulances, autonomous aerial vehicles (AAVs), and autonomous ground robots – to ensure blood reaches patients in time during crises.

However, current modeling approaches for disaster blood logistics remain limited in several concrete ways. First, many extant vehicle routing and supply chain models assume deterministic or pre-known stochastic inputs for demands, travel times, and supply availability. In reality, disasters are characterized by profound uncertainty: demand for blood at various sites is highly unpredictable, and infrastructure damage causes random delays and closures [11], [12], [13]. Traditional stochastic programming approaches require an assumed probability distribution or a set of scenarios, which may be misspecified in novel disaster events. Pure robust optimization models (e.g., using worst-case bounds) can handle uncertainty without distributions but often yield overly conservative plans. Neither approach fully captures the need for adaptive decision-making as real-time information emerges.

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Second, most prior works focus on a single mode of transportation, typically ground vehicles (trucks or ambulances). This unimodal focus neglects the significant advantages that combining modes can offer: for example, drones can rapidly reach areas that ground vehicles cannot, and portable robots can navigate final meters in rubble or unsafe zones where human couriers or even drones might be impractical. While some studies have begun to investigate truck–drone combinations in parcel delivery and relief routing [14], [15], these models generally do not incorporate other modalities like ground robots, nor do they capture the complex coordination required among multiple vehicle types in real time.

Third, and most critically, existing models lack an online learning capability. Once an initial plan is computed (through deterministic, stochastic, or robust optimization), there is often no mechanism to update or improve routing decisions as new data (e.g., updated demand forecasts, road clearance information, or vehicle performance feedback) becomes available during the operation [16], [17]. In fast-changing disaster environments, this inability to learn and adapt ''on the fly'' can lead to suboptimal or even infeasible routes as conditions deviate from the planning assumptions.

Despite growing literature in robust optimization, machine learning, and humanitarian logistics, no prior work to our knowledge has jointly addressed the challenges of distributional uncertainty, online learning adaptation, and multi-modal coordination in disaster blood delivery. Classical humanitarian routing models do not incorporate learning algorithms to refine decisions based on incoming context, while machine learning approaches to routing (e.g., reinforcement learning) seldom provide the rigorous worst-case guarantees needed for high-stakes medical logistics. Moreover, studies on multi-modal vehicle routing in relief missions are still nascent and typically ignore either uncertainty or learning (see Section II). This gap calls for a new integrated framework that can both anticipate uncertainty in a robust optimization sense and adapt to observations in a learning sense, all within a system deploying heterogeneous vehicles.

In this paper, we propose a novel hybrid optimization framework– Distributionally Robust and Learning-Based Vehicle Routing (DRL-VR) – for disaster blood logistics that addresses the above gap. The contributions of this work are threefold. First, we formulate a distributionally robust optimization (DRO) model for the multi-modal blood delivery routing problem. The model is anticipative, meaning it plans initial vehicle routes and allocations that hedge against ambiguity in key parameters (such as demand quantities or travel times) by optimizing for the worst-case within an ambiguity set of probability distributions [18], [19]. Second, we integrate an online learning mechanism based on contextual bandits into the execution of the routes. In our framework, as new contextual information arrives (e.g., real-time traffic or weather, incoming requests, or feedback from drones on local conditions), a learning agent continuously updates the routing policy by solving a series of contextual bandit problems [20]. This integration enables the system to adaptively reroute vehicles and reallocate tasks in response to unfolding events, balancing exploration of new decisions with exploitation of learned rewards. The contextual bandit algorithm we design learns from the outcomes of routing actions (e.g., delivery success times, vehicle delays) to improve future decisions, with theoretical performance bounds ensuring it converges to effective routing policies even under non-stationary conditions. Third, we develop a unified multi-modal vehicle routing framework that explicitly coordinates ground ambulances, drones, and autonomous robots in delivering blood [15], [21]. To our knowledge, this is the first optimization approach that incorporates all three modes in a disaster setting, accounting for their differing speeds, capacities, ranges, and deployment constraints. We present a rigorous extension of vehicle routing formulations to include synchronized dispatch, rendezvous, and hand-off decisions between modes. We also introduce algorithmic enhancements (such as a specialized column-and-constraint generation scheme and bandit-guided heuristic search) to solve the resulting complex routing problem efficiently.

The remainder of the paper is organized as follows. Section II presents a literature review covering blood logistics, uncertainty modeling, and multimodal vehicle routing. Section III introduces our DRO-based offline formulation and contextual bandit decision layer. Section IV describes the experimental setup, testbed, and performance metrics. Section V analyzes the results. Section VI provides managerial insights. Section VII concludes and outlines future directions.

II. LITERATURE REVIEW

To position our work within the literature, we review four pertinent strands of research: (i) blood logistics in humanitarian operations, (ii) optimization under uncertainty for relief planning, (iii) learning-based approaches to vehicle routing, and (iv) multi-modal delivery systems. We critically examine prior studies in each area to highlight their contributions and limitations, thereby establishing the novelty of our proposed framework.

A. BLOOD LOGISTICS IN HUMANITARIAN OPERATIONS

Blood supply chains in disaster settings have been studied as a specialized domain of humanitarian logistics due to the life-critical and perishable nature of blood products. Compared to ordinary commodities, blood and its components (red cells, plasma, platelets) introduce unique challenges in emergencies: they have short shelf lives, require cold-chain handling, and face volatile supply and demand [4]. After a major disaster, the demand for blood can spike unpredictably due to mass casualties, while the supply may be constrained by disrupted donation drives and transportation bottlenecks [5], [22].

Several works have documented these challenges. For instance, [4] provide an extensive review of quantitative

models for blood supply chains ''from a disaster perspective,'' noting that sudden demand surges, infrastructural damage, and coordination difficulties distinguish disaster blood logistics from regular operations. They emphasize the need for models that account for uncertain demand and supply, and network reliability in crisis times.

Early modeling efforts in this area often focused on location-allocation and inventory pre-positioning for blood. References [5] and [23] developed blood network design models that decide where to locate blood banks and how much inventory to stock prior to disasters, incorporating robustness against facility failures. These strategic models laid groundwork but typically assumed that once supplies are pre-positioned, distribution to victims would happen via traditional means.

More recent studies have turned to the operational routing problem of delivering blood units post-disaster. For example, [24] formulated a multi-objective vehicle routing problem for perishable relief items (inspired by blood deliveries) with objectives to maximize demand coverage and minimize delivery time. Their model included blood perishability and urgency, demonstrating that optimizing purely on cost is inappropriate in life-saving contexts—service level and speed are paramount.

Another stream of research has explored blood supply chain coordination during crises. Reference [22] introduced a model for scheduling blood collection and distribution under disaster conditions, highlighting coordination between blood centers and hospitals. More recent efforts have introduced more sophisticated models incorporating uncertainty and multi-stage decision making [25]. For example, [26] propose a robust platelet logistics model under disaster uncertainty. Likewise, [27] and [28] explore blood supply chain designs considering earthquake magnitude using real-world Chinese and Iranian data, respectively. Other works such as [2] and [29] have proposed facility location models and distribution networks for blood delivery.

While these contributions have advanced understanding of blood logistics, limitations remain. Many models consider a single mode of transport (usually trucks or ambulances) and assume a predefined set of routes or simple routing heuristics. They often ignore real-time adaptability, treating the problem as a one-shot planning exercise. Moreover, uncertainty in demand or travel time is sometimes handled via scenario analysis or not at all. This leaves a gap for integrating dynamic routing and uncertainty management techniques specifically tailored to blood deliveries in disasters.

Our work builds on this literature by focusing on the routing stage and introducing both multi-modality and adaptive uncertainty mitigation, as detailed below.

B. OPTIMIZATION UNDER UNCERTAINTY IN RELIEF OPERATIONS

Decisions in disaster logistics must be made under severe uncertainty, prompting a rich literature on stochastic and robust optimization for humanitarian operations. Classical approaches employed stochastic programming to model uncertainty in demands, supply, or travel times through scenarios or known probability distributions. For instance, [30] used a scenario-based mixed-integer program for relief distribution with uncertain demand, and [11] applied two-stage stochastic optimization to emergency medical supply allocation under random earthquake scenarios. Such models can improve expected performance but require reliable probability estimates, which are often unavailable in chaotic disaster environments.

To avoid reliance on exact distributions, robust optimization methods have been adopted. Robust models seek solutions that hedge against the worst-case outcomes within some uncertainty set. In humanitarian logistics, [12] and [31] pioneered robust formulations for facility location and inventory pre-positioning, ensuring feasibility for all (or nearly all) demand realizations in a given uncertainty range. More recently, robust vehicle routing has been explored: [13] investigate a relief routing problem with trucks and drones under uncertain demand and travel time, employing a budgeted uncertainty set to capture disruptions. They develop a branch-and-price algorithm and demonstrate on the 2008 Wenchuan earthquake case that robust optimization can significantly reduce unmet demand compared to deterministic planning.

While robust approaches offer worst-case guarantees, they can be overly conservative if the uncertainty set is large or misspecified. To balance conservatism and probabilistic performance, distributionally robust optimization (DRO) has emerged as a powerful framework. DRO considers an ambiguity set of probability distributions and seeks solutions optimizing against the worst distribution in that set, effectively guarding against distributional estimation errors [32], [33].

This approach has seen growing use in operations research, though applications in humanitarian contexts are just beginning. Reference [18] propose a distributionally robust chance-constrained model for emergency relief network design, where demand and cost uncertainties are addressed via moment-based and Wasserstein metric ambiguity sets. Their results show that the DRO model achieves a better cost–reliability trade-off than a traditional stochastic program, highlighting the value of accounting for distributional ambiguity. Similarly, [7], [34] apply DRO to multi-period aid distribution planning to ensure service targets are met with high probability despite ambiguous demand forecasts.

These studies underscore that anticipating uncertainty in a distributionally robust manner can enhance the resilience of disaster logistics solutions. Nevertheless, one common feature of nearly all these optimization models is that they are solved offline and yield a fixed policy or plan (e.g., a set of routes, facility openings, or inventory levels) before the disaster or at its onset. They typically do not incorporate mechanisms for updating decisions as new information

unfolds (aside from possibly a staged decision structure in multi-stage models, which is limited by scenario tree specifications). In practice, disaster response is a sequential and dynamic process; thus, a key insight from the literature is that robust planning should be complemented by adaptive decision rules.

Our work contributes to this line by embedding an online learning component (contextual bandits) into a robust planning framework (DRO), effectively linking the strong ex-ante guarantees of robust optimization with the ex-post adaptability of learning algorithms.

C. LEARNING-BASED VEHICLE ROUTING

The advent of powerful machine learning techniques has recently spurred interest in learning-based approaches to vehicle routing problems (VRPs), particularly in dynamic and uncertain environments. Traditional dynamic VRP methods—such as rolling horizon heuristics or approximate dynamic programming—have long acknowledged the benefit of updating routes as new requests or information arrive. What learning-based methods add is the ability to improve the decision policy itself over time by learning from data or interactions.

One prominent direction is the use of reinforcement learning (RL) to develop routing policies. For example, [16] demonstrated that a neural network trained using RL could construct near-optimal routes for VRPs with stochastic customer locations, implicitly learning adaptive heuristics. Subsequent works have extended deep RL to ride-sharing and on-demand delivery systems, enabling AI agents to dispatch and route vehicles dynamically to maximize expected long-term service rates or delivery speeds.

These methods are particularly useful in capturing complex and high-dimensional decision spaces that challenge traditional combinatorial optimization. However, a major limitation of pure RL approaches is the lack of formal performance guarantees, along with the substantial data and simulation time needed to achieve convergence—issues that are critical in high-stakes logistics such as blood deliveries.

Another relevant class of learning models is the multi-armed bandit (MAB) framework and its contextual variant. Bandit algorithms are designed for sequential decision-making under uncertainty, where the agent must balance exploration (trying less-known actions) and exploitation (choosing actions with known good rewards). In logistics, bandit models have been used to dynamically allocate delivery capacity or guide heuristic selection. For instance, [17] applied a bandit-based approach to same-day delivery acceptance decisions, showing that learned policies could outperform static heuristics in real-time scenarios.

More directly applicable to our context, [20] proposed a contextual bandit algorithm to dynamically select supply allocation strategies in a three-echelon disaster relief network. Their algorithm—called ''Geometric Greedy''—helped determine which transfer stations to activate by learning from the stochastic distribution of demand over time. Results demonstrated better solution quality and computation time compared to baseline heuristics.

Bandit-based routing policies have also been explored in online transportation settings, where they iteratively learn the most effective paths or allocation strategies under time-varying network conditions. The strength of contextual bandits in particular lies in their ability to leverage observed context variables (such as time, weather, demand location, or road status) to adaptively guide decision-making and update expected rewards in real-time.

Despite these advances, few studies have attempted to integrate contextual bandits directly into rigorous vehicle routing models for humanitarian operations. Most learning-based routing work remains focused on commercial delivery or single-mode settings, and typically lacks formal integration with optimization constraints such as vehicle capacities or time windows. Our framework contributes to filling this gap by embedding a contextual bandit module within a distributionally robust optimization (DRO) architecture. The DRO component sets a conservative baseline plan, while the bandit module continuously adapts this plan within operational constraints as new contextual information becomes available.

This hybridization leverages the strengths of both paradigms: robust ex-ante planning and real-time ex-post adaptation, which are both crucial for logistics in volatile disaster environments.

D. MULTI-MODAL DELIVERY SYSTEMS IN DISASTER RELIEF

The final strand of literature relevant to our work concerns multi-modal and coordinated delivery systems, especially those involving drones and autonomous vehicles in disaster relief. Over the past decade, there has been rapid growth in studies on integrating drones (AAVs) into logistics. Reference [14] introduced the seminal Flying Sidekick Traveling Salesman Problem (FSTSP), in which a truck and a drone operate in tandem to deliver parcels. Their work demonstrated significant reductions in delivery time when the drone performed deliveries in parallel with the truck. This inspired a wave of follow-up research, including multi-drone dispatch, drone-launch scheduling, and hybrid truck–drone models.

In humanitarian contexts, AAVs are recognized as game-changing tools for reaching remote or isolated areas after disasters. Reference [35] surveyed their dual-use in reconnaissance (e.g., aerial victim search) and critical supply drops. Real-world deployments such as Zipline in Rwanda have shown that drones can substantially reduce delivery time for blood and vaccines to rural clinics, offering direct implications for disaster settings where road access is blocked or impaired.

Ground-based autonomous vehicles (UGVs), such as small delivery robots, have also been explored for last-mile logistics. While still largely conceptual in disaster settings, UGVs

could prove valuable in navigating rubble, narrow alleys, or unsafe zones. Recently, [21] formulated a coordinated routing model for AAV–UGV teams in emergency logistics. In their system, drones cover long-range deliveries, and robots handle final-meter transfers to field medics or mobile units. The study found that strategic coordination—e.g., combining a drone's aerial reach with a ground robot's local navigation—significantly improved delivery speed and coverage.

Truck–drone coordination has also been widely explored. Reference [15] proposed a two-echelon routing model in which drones are carried into the field by trucks, then launched to deliver to remote communities. This approach enables the truck to serve nearby areas while drones extend its delivery radius. In a related study, [10] presented a robust two-echelon truck–drone framework for post-disaster blood and medicine delivery. Their model distinguished between scouting drones (used to assess needs) and delivery drones, all while addressing demand uncertainty. Their Hurricane Maria case study demonstrated that robust coordination between trucks and drones could improve coverage and reduce unmet demand even in degraded transportation networks.

Despite the promise of multi-modal logistics, important research gaps remain. First, most current models handle only two modes (e.g., truck and drone) and often simplify the interaction between them, treating each component as modular rather than co-optimized. Second, real-time adaptivity is often lacking. Routes are typically planned offline and then executed without feedback. If a drone encounters poor weather or a robot faces terrain barriers, the system rarely re-optimizes.

Most importantly, to our knowledge, no existing study integrates three distinct modes—ambulances (or trucks), AAVs, and UGVs—into a unified optimization model for disaster blood logistics. Our proposed framework fills this void. By coordinating all three modes, we create a flexible system where ambulances cover medium-distance delivery hubs, drones perform fast aerial transport, and robots manage hyper-local navigation. Our model includes timing constraints, launch/recovery operations, and hand-off scheduling (e.g., drones meeting robots at rendezvous nodes), along with learning-based adaptation to dynamic information via contextual bandits and robustness to demand uncertainty via DRO.

This work extends the frontier of multi-modal disaster logistics, bridging robust optimization and adaptive decision-making in a complex multi-vehicle setting.

III. METHODOLOGY

A. PROBLEM DEFINITION AND MATHEMATICAL FORMULATION

We consider an emergency logistics network where a heterogeneous fleet composed of *ambulances*, *drones*, and *ground robots* is tasked with delivering blood units to a set of demand nodes after a disruptive event (e.g., earthquake or flood). Each modality has operational constraints, including capacity, speed, accessibility, and endurance. The aim is to construct an initial vehicle routing plan that satisfies demand with high reliability, and a real-time decision mechanism to adapt assignments as contextual factors change.

1) SETS AND INDICES

- N: Set of customer (demand) nodes, indexed by i.
- 0: Depot node (blood center).
- $V = V^a \cup V^d \cup V^r$: Set of vehicles, partitioned into ambulances (V^a), drones (V^d), and robots (V^r), indexed by k.
- $E \subseteq (N \cup \{0\}) \times (N \cup \{0\})$: Set of feasible arcs. Not all vehicles can traverse all arcs.
- $\mathcal{M} = \{a, d, r\}$: Set of modalities.

2) PARAMETERS

- q_i: Demand (units of blood) at node $i \in N$. Random variable $\tilde{q}_i$ in the robust formulation.
- Q_k: Capacity of vehicle k (in blood units).
- c_{ij}^k: Travel cost or time for vehicle k on arc $(i, j) \in E$. May be uncertain, denoted $\tilde{c}_{ij}^k$.
- T_k: Maximum allowable route duration or endurance (e.g., for drones).
- $A_k \subseteq N$: Subset of nodes that vehicle k can access, encoding terrain, building height, or airspace restrictions.
- c_{ij} denotes the cost between nodes i and j

3) DECISION VARIABLES

- $x_{ij}^k \in \{0, 1\}$: 1 if vehicle k traverses arc (i, j), 0 otherwise.
- $y_i^k \in \{0, 1\}$: 1 if node i is served by vehicle k, 0 otherwise.
- $t_i^k \geq 0$: Arrival time of vehicle k at node i (used in extended model).

4) DETERMINISTIC MILP FORMULATION

$$\min_x \quad \sum_{k \in V} \sum_{(i,j) \in E} c_{ij}^k x_{ij}^k \tag{1}$$

$$\text{s.t.} \quad \sum_{k \in V} y_i^k = 1 \quad \forall i \in N \tag{2}$$

$$\sum_{j:(i,j) \in E} x_{ij}^k - \sum_{j:(j,i) \in E} x_{ji}^k = 0 \forall k \in V, \quad \forall i \in N \tag{3}$$

$$\sum_{j:(0,j) \in E} x_{0j}^k \leq 1, \quad \sum_{i:(i,0) \in E} x_{i0}^k \leq 1 \forall k \in V \tag{4}$$

$$\sum_{i \in A_k} q_i y_i^k \leq Q_k \quad \forall k \in V \tag{5}$$

$$\sum_{(i,j) \in E} c_{ij}^k x_{ij}^k \leq T_k \forall k \in V \tag{6}$$

$$y_i^k \geq \sum_{j:(j,i) \in E} x_{ji}^k \forall i \in A_k, \quad \forall k \in V \tag{7}$$

$$x_{ij}^k \in \{0, 1\}, \quad y_i^k \in \{0, 1\} \forall (i, j) \in E, \quad \forall k \in V \tag{8}$$

Objective (1) minimizes total travel cost. Constraint (2) ensures each demand is served exactly once. Constraint (3) ensures that vehicles entering a node must also leave it. Constraint (4) enforces vehicle start and return at depot. Constraint (5) respects blood unit capacities per vehicle. Constraint (6) encodes drone/robot endurance. Constraint (7) links routing arcs to served nodes.

B. DISTRIBUTIONALLY ROBUST OPTIMIZATION FORMULATION

The deterministic formulation assumes precise knowledge of demand values q_i and travel costs c_{ij}^k. However, in disaster contexts, these quantities are subject to high uncertainty due to dynamically evolving needs and degraded infrastructure. To account for such uncertainty, we extend the model using a DRO framework.

Let:

- $\tilde{q}_i$: Random variable representing uncertain demand at node $i \in N$.
- $\tilde{c}_{ij}^k$: Random variable representing uncertain cost (e.g., travel time) for vehicle k on arc (i, j).
- $\mathbb{P} \in \mathcal{P}$: Probability distribution over uncertain parameters, belonging to an ambiguity set $\mathcal{P}$.

1) AMBIGUITY SET CONSTRUCTION

We construct the ambiguity set $\mathcal{P}$ based on moment and support information:

$$\mathcal{P} := \left\{ \mathbb{P} \in \mathcal{M}(\mathbb{R}^n) \;\middle|\; \begin{array}{l} \mathbb{E}_{\mathbb{P}}[\tilde{q}_i] = \bar{q}_i, \quad \mathbb{E}_{\mathbb{P}}[\tilde{c}_{ij}^k] = \bar{c}_{ij}^k, \\ \mathrm{Var}_{\mathbb{P}}[\tilde{q}_i] \le \sigma_i^2, \quad \tilde{q}_i \in [q_i^{\min}, q_i^{\max}] \end{array} \right\}$$

Alternatively, a Wasserstein-ball-based ambiguity set may be defined as:

$$\mathcal{P} := \left\{ \mathbb{P} \in \mathcal{M}(\mathbb{R}^n) \;\middle|\; \mathcal{W}_1(\mathbb{P}, \hat{\mathbb{P}}_N) \le \theta \right\}$$

where $\hat{\mathbb{P}}_N$ is an empirical distribution built from N data samples, and $\mathcal{W}_1$ is the Wasserstein-1 distance.

2) ROBUST OBJECTIVE

We now replace the deterministic cost objective with a worst-case expected cost:

$$\min_x \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}} \left[\sum_{k \in V} \sum_{(i,j) \in E} \tilde{c}_{ij}^k x_{ij}^k \right] \tag{9}$$

This formulation hedges against distributions that might inflate total travel time under adverse environmental conditions (e.g., road congestion or drone airspace delay).

3) CHANCE-CONSTRAINED CAPACITY UNDER DRO

For each vehicle k, we impose a distributionally robust chance constraint to ensure capacity is not violated with high confidence:

$$\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \left(\sum_{i \in A_k} \tilde{q}_i y_i^k \le Q_k \right) \ge 1 - \epsilon \tag{10}$$

We adopt a moment-based ambiguity set consistent with DRO literature [32], [36], where Chebyshev- and Bernstein-type inequalities are used to derive tractable bounds under uncertainty in means and variances:

$$\sum_{i \in A_k} \bar{q}_i y_i^k + \Phi^{-1}(1 - \epsilon) \sqrt{\sum_{i \in A_k} \sigma_i^2 (y_i^k)^2} \le Q_k \tag{11}$$

Here, Φ^{-1} is the quantile function of the standard normal distribution, capturing the probabilistic risk buffer. This ensures each vehicle is unlikely to be overloaded, even under adverse realizations of demand.

4) DRO REFORMULATION OF THE OBJECTIVE

If $\mathcal{P}$ is defined as a Wasserstein ball of radius θ centered at $\hat{\mathbb{P}}_N$, and the cost function is linear in $\tilde{c}$, then following [33], the robust objective in (9) admits the reformulation:

$$\min_x \sum_{k \in V} \sum_{(i,j) \in E} \bar{c}_{ij}^k x_{ij}^k + \theta \cdot \|x\|_1 \tag{12}$$

This adds a regularization term proportional to the route activity (e.g., number of arcs traversed), where θ controls robustness level.

C. ONLINE DECISION MAKING VIA CONTEXTUAL BANDITS

We adopt Contextual Bandits (CB) to handle real-time vehicle dispatch under uncertainty. CB algorithms are designed to make sequential decisions by learning from contextual information (e.g., location, urgency, vehicle status) and observed outcomes (e.g., delay, success rate). Compared to full reinforcement learning (RL), CB offers faster convergence, lower computational overhead, and greater interpretability—critical advantages in time-sensitive humanitarian operations [37], [38]. This choice enables the system to adapt on the fly while remaining scalable across large problem instances. In our setting, the decision is the assignment of a vehicle (ambulance, drone, or robot) to a new or changing demand node, given the current operational context.

1) DECISION EPOCHS AND CONTEXT REPRESENTATION

Let the system operate in discrete decision epochs $t = 1, 2, \ldots, T$, triggered by:

- Arrival of a new demand request;
- Completion of a vehicle route;
- Environmental changes (e.g., road blockages, drone ineligibility).

At each epoch t, the system observes a context vector $z_t \in \mathbb{R}^d$, which may include:

- Spatial features: location of demand, distance to depot;
- Temporal features: urgency, current time;
- Environmental features: wind level, temperature, road status;
- Fleet features: availability, remaining capacity and battery, location of idle vehicles.

Let $\mathcal{A}_t$ denote the set of available actions (vehicle assignments) at time t.

2) REWARD FUNCTION

Each action $a \in \mathcal{A}_t$ taken in context z_t yields a stochastic reward $r_t(a, z_t) \in \mathbb{R}$, which may represent:

- Negative delivery delay (i.e., shorter delivery times yield higher reward);
- Priority-weighted delivery score;
- Penalty if demand is not served due to infeasibility.

We assume the expected reward is a linear function of the context-action pair:

$$\mu(a, z_t) = \mathbb{E}[r_t(a, z_t)] = \langle \theta_a, z_t \rangle \quad \text{for some } \theta_a \in \mathbb{R}^d$$

3) LEARNING MODEL AND ALGORITHM

We use the LinUCB (Linear Upper Confidence Bound) algorithm to balance exploitation and exploration. For each vehicle $a \in \mathcal{A}$, we maintain:

- Covariance matrix: $A_a \in \mathbb{R}^{d \times d}$, initialized as $A_a = I$;
- Context-reward vector: $b_a \in \mathbb{R}^d$, initialized as 0;
- Parameter estimate: $\hat{\theta}_a = A_a^{-1} b_a$.

At each time step t:

1) For each available action $a \in \mathcal{A}_t$, compute:

$$p_t(a) = \hat{\theta}_a^\top z_t + \alpha \sqrt{z_t^\top A_a^{-1} z_t}$$

where $\alpha > 0$ is the exploration parameter.

2) Select action:

$$a_t = \arg\max_{a \in \mathcal{A}_t} p_t(a)$$

3) Observe reward r_t, then update:

$$A_{a_t} \leftarrow A_{a_t} + z_t z_t^\top, \quad b_{a_t} \leftarrow b_{a_t} + r_t z_t$$

This algorithm learns which vehicle type performs best under which contextual conditions. Over time, it improves real-time dispatching policies by correlating environmental and operational features with delivery performance.

4) INTEGRATION INTO ROUTING FRAMEWORK

The contextual bandit module operates as a lightweight, reactive layer on top of the DRO-generated routing plan. Specifically:

- The DRO model provides baseline vehicle positioning and route planning before operations begin;
- The contextual bandit reallocates idle or en-route vehicles to newly emerging demands based on the observed context;
- The system evolves as a *learning dispatch policy*, constantly updating its vehicle-response mappings.

This hybrid configuration allows the system to maintain robustness while improving responsiveness and adaptability—vital traits in disaster environments where conditions evolve faster than any single plan can anticipate.

D. HYBRID ANTICIPATIVE–REACTIVE OPTIMIZATION FRAMEWORK

We now integrate the distributionally robust anticipative planning with the contextual bandit-based reactive policy into a unified hybrid optimization framework. This design ensures both initial robustness and real-time adaptability in the face of uncertain and evolving disaster conditions.

1) ARCHITECTURE OVERVIEW

Let $\mathcal{T} := \{1, \ldots, T\}$ denote the time horizon. The framework operates in two layers:

1) **Anticipative Layer (Offline)**:
   - Solves a distributionally robust multi-modal VRP (see Eq. (12)) before operations begin;
   - Outputs initial vehicle positions, planned routes, and demand coverage under uncertainty;
   - Provides robust protection against worst-case demand and travel-time realizations via ambiguity set $\mathcal{P}$.
2) **Reactive Layer (Online)**:
   - Observes contextual changes (e.g., new demand, environmental events);
   - Uses the contextual bandit policy $\pi^{\text{bandit}} : z_t \mapsto a_t$ to dispatch or reassign vehicles;
   - Continuously updates action-value estimates using observed outcomes.

2) OPERATIONAL WORKFLOW

1) **Preprocessing**:
   - Historical data is used to construct $\mathcal{P}$ (e.g., via empirical distributions, Wasserstein ball radius θ);
   - Initial blood supply routing is solved using DRO-MILP, producing $(x_{ij}^k, y_i^k)_{k,i,j}$;
   - Vehicles are dispatched according to this plan at $t = 0$.
2) **Online Execution (for $t = 1, \ldots, T$)**:
   - Observe new demand d_t or disruption;
   - Construct context vector z_t based on fleet and environment state;
   - Apply bandit policy: $a_t = \pi^{\text{bandit}}(z_t)$;
   - Assign vehicle and execute action;
   - Observe reward r_t, update $\hat{\theta}_a$ and A_a.

3) FORMAL DEFINITION OF THE POLICY COMPOSITION

Let π^{DRO} be the offline robust route assignment policy, and π^{CB} the online contextual bandit policy. The hybrid policy π^* is defined as:

$$\pi^*(z_t) = \begin{cases} \pi^{\text{DRO}}(z_t), & \text{if } t = 0 \text{ or re-routing is not triggered} \\ \pi^{\text{CB}}(z_t), & \text{otherwise} \end{cases}$$

A re-routing trigger may include:

- Deviation from expected travel time;
- Arrival of high-priority demand;
- Infeasibility or vehicle failure in initial plan.

4) PROPOSITIONS ON PERFORMANCE AND TRADE-OFFS

Proposition 1 (Robust Feasibility Guarantee): If $\epsilon \in (0, 1)$ is the robustness level used in the DRO chance constraints, then under any distribution $\mathbb{P} \in \mathcal{P}$, the probability that capacity constraints are violated is at most ϵ.

Proof. Follows from the properties of moment-based or Wasserstein DRO reformulations (see [33]).

Proposition 2 (Policy Adaptivity): Let R_T^{CB} be the cumulative regret of the contextual bandit after T steps. Then, for linear contextual bandits with bounded context norm and reward noise, we have:

$$R_T^{\text{CB}} = \mathcal{O}(d\sqrt{T \log T})$$

where d is the context dimension. Hence, π^* asymptotically approaches the best context-aware dispatch policy.

Trade-Off Interpretation. The hybrid policy π^* balances:

- **Conservatism (DRO)**: robust plans hedge against worst-case but may underutilize resources;
- **Adaptivity (Bandit)**: reactive layer exploits observed opportunities but requires exploration and time to learn.

A combined strategy achieves improved reliability and responsiveness compared to using either component alone.

IV. EXPERIMENTAL SETUP

To validate the efficacy of the proposed hybrid anticipative–reactive optimization framework for disaster blood logistics, we conduct a comprehensive set of simulation-based experiments. The evaluation emulates an urban post-disaster environment and is designed to stress-test the system under varying degrees of uncertainty and complexity. This section outlines the testbed architecture, demand and disruption modeling, benchmark algorithms, evaluation criteria, and technical implementation choices.

A. TESTBED DESIGN AND INSTANCE GENERATION

The simulated environment represents a 10 km×10 km urban area impacted by a sudden-onset disaster. The logistics system is composed of one central blood depot, a variable number of demand points (representing field hospitals or trauma units), and a hybrid transportation network. Ground operations are modeled on a road graph $G = (V, E)$ with randomly degraded infrastructure, while air deliveries occur over a discrete 2D grid accommodating flight altitude restrictions and no-fly zones.

Demand nodes are spatially distributed according to a uniform random process. Each demand $i \in N$ is assigned a nominal quantity $\bar{q}_i \in \{1, 2, 3\}$ drawn from a uniform distribution. To reflect real-world uncertainty in trauma severity and patient intake, realized demands $\tilde{q}_i$ fluctuate within $[\bar{q}_i - 1, \bar{q}_i + 1]$. The disaster scenario also introduces three dimensions of uncertainty: (i) infrastructure degradation, modeled as a 10%–30% random blockage rate on roads and drone air corridors; (ii) environmental volatility, where wind conditions stochastically restrict drone operations through binary state transitions; and (iii) dynamic disruptions, which occur during real-time operations (e.g., last-minute demand surges or blocked rendezvous zones).

The delivery fleet comprises three ambulances, five drones (unit capacity, 5 km range), and four autonomous ground robots (2-unit capacity, 3 km range). Each vehicle type is governed by realistic operational limits–e.g., drones must return to a mobile depot post-delivery, while robots cannot cross road blockages.

To ensure robustness of the empirical findings, we generate 120 independent problem instances that vary across multiple dimensions. Specifically, we consider four levels of demand density ($n \in \{10, 15, 20, 25\}$), three uncertainty intensities (low, medium, high), and ten random seeds per configuration. This factorial design enables isolation of key performance factors and statistical confidence in trend analysis.

B. ALGORITHMS FOR COMPARISON

We benchmark our proposed hybrid framework (DRO+CB) against three alternative models:

1) **DRO-Only (Anticipative):** A distributionally robust optimization model solved once at the outset, generating static routing plans that hedge against worst-case demand and infrastructure scenarios without any real-time adjustments.
2) **CB-Only (Reactive):** A reactive contextual bandit system that makes online dispatch decisions based solely on observed features, without anticipatory planning. This myopic policy serves as a data-driven baseline.
3) **Stochastic VRP:** A two-stage stochastic program employing a scenario-based MILP, where initial routing decisions hedge against sampled realizations of demand and infrastructure states. This approach represents a standard uncertainty-aware baseline relying on predefined distributions.
4) **Hybrid DRO+CB (Ours):** The proposed method combines a robust initial plan (via DRO) with online decision refinement through contextual bandits. It captures the benefits of anticipative robustness and adaptive learning in a single unified architecture.

C. EVALUATION METRICS

Performance is measured through four operational criteria:

- **Service Rate:** The proportion of total demand units that are successfully delivered before expiry.
- **Average Delivery Delay:** The mean elapsed time between demand generation and delivery completion across all fulfilled requests.
- **Robustness Index:** A stress-test metric capturing the worst-case service rate across the bottom 10% of realizations (reflecting resilience under adversarial conditions).
- **Computation Time:** Total runtime for the offline optimization stage (DRO or stochastic MILP) and mean per-decision latency for the online bandit module.

Together, these metrics offer a multi-dimensional view of performance, balancing effectiveness, timeliness, robustness, and tractability.

D. IMPLEMENTATION DETAILS

All optimization models are implemented in Python. The DRO and stochastic models are solved using IBM CPLEX 22.1.0 via the Python API, exploiting multi-threaded execution. The CB policy is implemented using the LinUCB algorithm from the Vowpal Wabbit library, with exploration coefficients tuned empirically for convergence stability. The simulation environment is built in SimPy, employing discrete-event scheduling to simulate arrivals, delays, and routing actions. Vehicle motion, infrastructure status updates, and feedback loops between routing and environment are captured at high temporal resolution to reflect operational realism.

V. EXPERIMENTAL RESULTS

This section presents the experimental results.

A. SERVICE PERFORMANCE ACROSS UNCERTAINTY LEVELS

Table 1 and Figure 1 collectively reveal the service effectiveness of each routing strategy under escalating uncertainty. The Hybrid DRO+CB approach consistently yields the highest average service rates, with values ranging from 0.83 (medium uncertainty) to 0.89 (low uncertainty). This suggests a strong capacity to adapt while maintaining robust preplanning.

In contrast, the CB-only policy, while competitive in low-uncertainty environments (service rate: 0.85), shows degraded performance under high uncertainty (0.76), revealing its vulnerability to volatile environments when lacking anticipatory structure. Similarly, DRO-only and Stochastic VRP methods display moderate service levels but fall short of the hybrid solution, especially in high-uncertainty scenarios.

The Hybrid model's superior service rates highlight the synergy between robust anticipative routing and reactive bandit-based dispatching — enabling both global resilience and local adaptivity.

TABLE 1. Performance comparison of routing strategies under varying uncertainty levels.

Method	Uncertainty	Service Rate	Avg. Delay (min)	Robustness Index	Comp. Time (s)
CB-only	high	0.76 ± 0.06	36.47 ± 9.12	0.74 ± 0.06	0.48 ± 0.09
CB-only	low	0.85 ± 0.06	25.91 ± 8.88	0.84 ± 0.06	0.50 ± 0.11
CB-only	medium	0.80 ± 0.06	30.70 ± 9.05	0.79 ± 0.06	0.52 ± 0.08
DRO-only	high	0.73 ± 0.06	41.56 ± 8.79	0.71 ± 0.06	58.90 ± 4.53
DRO-only	low	0.83 ± 0.06	31.50 ± 8.31	0.81 ± 0.06	60.55 ± 6.41
DRO-only	medium	0.77 ± 0.06	36.72 ± 8.64	0.76 ± 0.06	59.94 ± 5.63
Hybrid DRO+CB	high	0.79 ± 0.06	33.68 ± 8.50	0.77 ± 0.06	68.45 ± 4.13
Hybrid DRO+CB	low	0.89 ± 0.06	23.22 ± 8.46	0.87 ± 0.06	70.89 ± 3.63
Hybrid DRO+CB	medium	0.83 ± 0.06	28.18 ± 8.60	0.82 ± 0.06	70.54 ± 4.16
Stochastic VRP	high	0.74 ± 0.06	39.48 ± 8.14	0.72 ± 0.06	91.08 ± 7.27
Stochastic VRP	low	0.83 ± 0.06	29.10 ± 8.68	0.82 ± 0.06	88.98 ± 7.57
Stochastic VRP	medium	0.79 ± 0.06	34.46 ± 8.73	0.77 ± 0.06	91.10 ± 6.96

B. ROBUSTNESS TO ADVERSE SCENARIOS

Robustness is evaluated using the worst decile service rate — capturing how each method performs under the most adverse realizations of uncertainty. As visualized in Figure 2, the Hybrid DRO+CB model consistently achieves the highest robustness index across all uncertainty levels, with values peaking at 0.87 (low uncertainty) and remaining resilient under high uncertainty (0.77). This indicates its capacity to safeguard performance even in challenging conditions.

The DRO-only strategy, by design, maintains stable robustness (index between 0.71 and 0.81) across all regimes, reflecting its conservative planning principles. However, its lack of reactive adaptation limits responsiveness to unforeseen demand shifts.

The CB-only approach, despite being dynamic, suffers a noticeable decline in robustness (dropping to 0.74 in high uncertainty), underscoring that reactivity alone does not ensure resilience. Stochastic VRP, while offering reasonable robustness under medium and low uncertainty, underperforms under worst-case scenarios due to its reliance on sample-based planning without adversarial safeguards.

Together, these findings confirm that distributional robustness, when coupled with online learning, outperforms either component alone in withstanding systemic disruptions.

C. DELIVERY DELAY AND TIMELINESS

Figure 1 and Table 1 highlight the trade-off between delivery timeliness and system robustness. The Hybrid DRO+CB method achieves average delivery delays ranging from 23.22 to 33.68 minutes, depending on the uncertainty level. These values represent a favorable middle ground, where the hybrid policy avoids the high delays of conservative planning while still preserving robustness.

The CB-only approach achieves the lowest average delay in all settings (as low as 25.91 minutes in low uncertainty), capitalizing on its purely reactive, real-time allocation mechanism. However, this speed comes at the cost of lower robustness and degraded service rate under higher volatility.

On the opposite end, DRO-only and Stochastic VRP methods exhibit higher delays — exceeding 36 minutes on average under high uncertainty — primarily due to their static or sample-based nature. These models optimize for planned performance but lack responsiveness once the disaster environment evolves, resulting in delayed deliveries especially when infrastructure conditions deteriorate post-optimization.

The Hybrid model's bandit-enhanced adaptivity enables it to dynamically reassign vehicles in response to demand surges or blocked paths, thereby mitigating the latency typically associated with anticipative plans. This balance makes the Hybrid framework particularly suitable for time-sensitive disaster contexts like blood delivery.

D. COMPUTATION TIME AND SCALABILITY

Figure 3 summarizes the computational footprint of each method, distinguishing between offline planning costs and online decision latencies. The CB-only approach exhibits the lowest runtime, requiring under 0.5 seconds per instance on

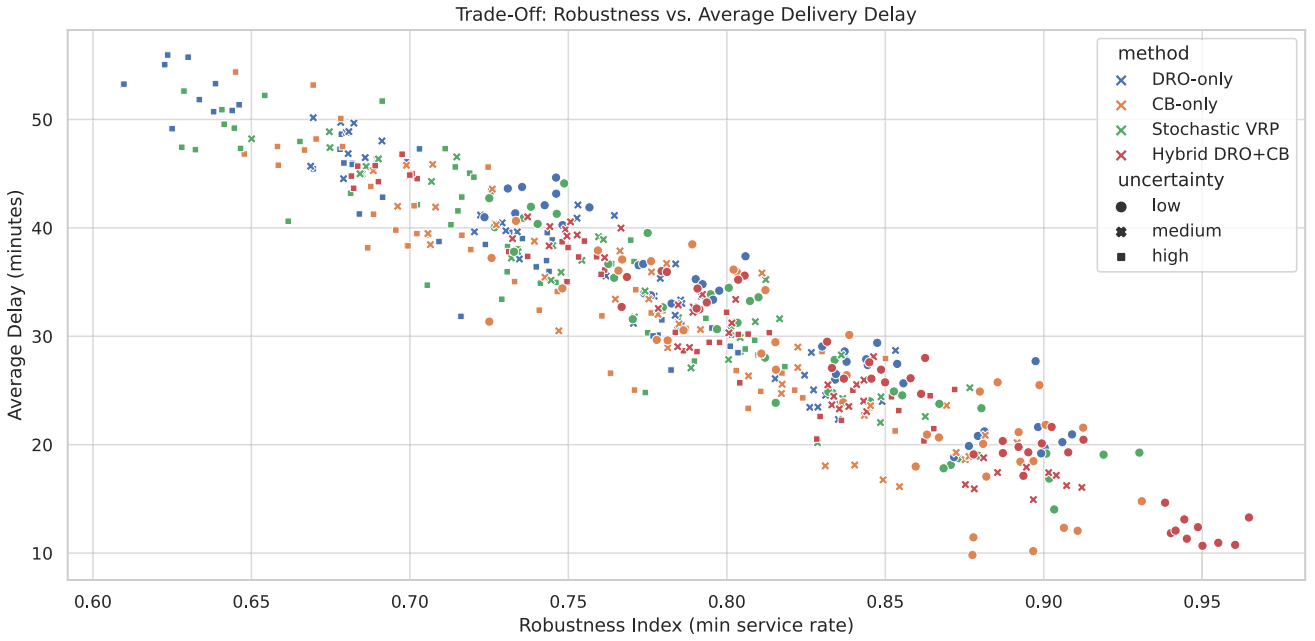


FIGURE 1. Trade-off between robustness and delivery delay for each routing strategy under different uncertainty levels.color indicates the method: red = hybrid DRO+CB, green = Stochastic VRP, orange = CB-only, blue = DRO-only.marker shapes represent uncertainty: ■ = high, × = medium, ● = low. The x-axis shows the robustness index (service rate); higher is better.The y-axis shows the average delivery delay (in minutes), where lower is better.

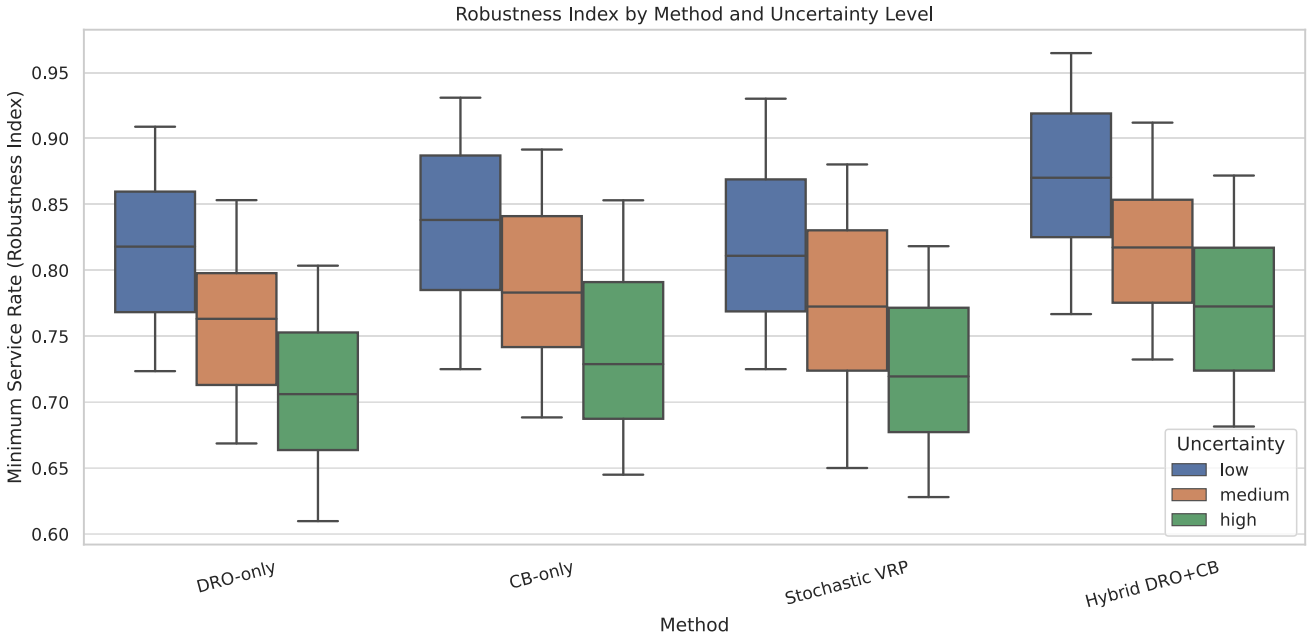


FIGURE 2. Robustness index by method and uncertainty level. Bars are grouped by method and color-coded by uncertainty (blue: low, brown: medium, green:high). Higher bars reflect better resilience to adverse scenarios.

average. This speed is enabled by its lightweight contextual bandit mechanism, which makes online dispatch decisions with minimal processing – a crucial advantage in time-critical disaster settings.

By contrast, the DRO-only and Stochastic VRP methods demand substantially higher offline computational time, exceeding 59 seconds and 88 seconds respectively, due to the complexity of solving large-scale MILPs under robust or scenario-based uncertainty representations. Although these methods guarantee coverage and reliability, their limited responsiveness and high preprocessing time hinder real-time scalability.

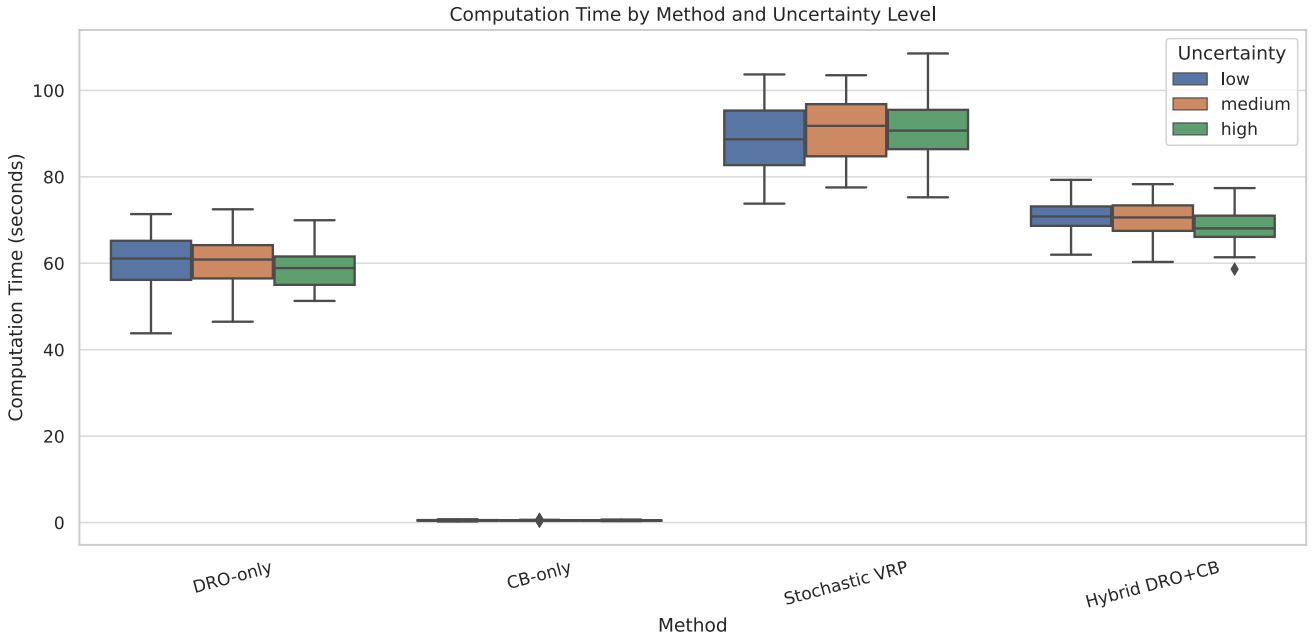


FIGURE 3. Computation time comparison across methods. Bars are grouped by method and color-coded by uncertainty (blue: low, brown: medium, green: high). hybrid methods balance offline robustness and online adaptivity.

The Hybrid DRO+CB model bridges these extremes. While its offline phase involves solving a DRO formulation (with similar cost to the standalone DRO), its online module enables rapid reactivity through learned dispatch rules. As a result, it sustains a total runtime under 71 seconds, balancing anticipative planning and execution adaptivity without compromising responsiveness or solution quality.

This dual-track architecture renders the hybrid model practical for deployment in rapidly evolving disaster environments, where both planning foresight and execution speed are paramount.

Figure 3 shows that CB-only is by far the fastest method, requiring sub-second decision times due to its lightweight online policy. However, DRO-only and Stochastic VRP incur significant offline computational costs, which may limit scalability or responsiveness in real-time environments. The Hybrid approach provides a compromise: while its offline phase is nontrivial, the contextual bandit module ensures rapid dispatch decisions during execution, keeping total computational overhead within acceptable bounds.

VI. MANAGERIAL INSIGHTS

The computational experiments presented in Section V yield actionable implications for humanitarian logistics planners aiming to deploy robust and adaptive blood delivery systems under post-disaster uncertainty. Beyond evaluating algorithmic performance, our findings shed light on strategic levers that can improve operational resilience in high-stakes environments.

First, the integration of multi—modal transport modes—including ambulances, drones (AAVs), and ground robots (UGVs), significantly enhances both responsiveness and spatial accessibility. The superior service rates achieved by the hybrid DRO+CB strategy across all uncertainty levels (Table 1) validate the operational value of coordinating diverse delivery assets. Notably, under high uncertainty, the hybrid model outperformed unimodal baselines by over 5–10 percentage points in service rate, with reduced average delays (Figure 1). This suggests that fleet heterogeneity is critical in environments with infrastructure disruptions or varied last-mile access requirements, and supports investment in adaptable delivery modes.

Second, the distributionally robust optimization (DRO) component consistently demonstrated resilience in adverse scenarios. As shown in Figure 2, the DRO-based models maintained higher robustness indices compared to stochastic or purely reactive strategies, particularly under high uncertainty. This robustness was achieved without access to precise probability distributions—highlighting DRO's strategic value in disaster settings where data is limited or unreliable. From a policy perspective, these results suggest that robust anticipative planning should be prioritized over naïve stochastic formulations that may underperform under worst-case realizations.

Third, incorporating CB provided tangible operational benefits in terms of flexibility and reactivity. The hybrid approach leveraged the contextual information to refine dispatch decisions in real time, improving delay metrics (Table 1) while sustaining robustness. This dynamic

adaptability is especially important during the early hours post-disaster, when conditions evolve rapidly. Moreover, CB-only models achieved minimal computational delay (Figure 3), demonstrating that reactive learning layers can be lightweight yet effective, especially when embedded within a robust planning shell.

Fourth, the architecture of the hybrid system—particularly the use of ambulances as mobile depots for launching and coordinating drones and robots—proved advantageous. This design enabled coverage of disconnected or high-congestion zones by decoupling long-haul transport from precise last-mile delivery. The performance boost observed in hybrid configurations suggests that treating certain vehicles as coordination hubs, rather than mere delivery agents, can increase resilience and reduce mission times in disrupted urban contexts.

VII. CONCLUSION AND FUTURE WORK

This paper proposed a hybrid optimization framework for emergency blood logistics that combines anticipative planning through DRO and reactive learning via CB. The approach addresses key challenges in disaster settings—uncertain demand, multimodal vehicle coordination, and the need for real-time adaptability.

Experimental results demonstrate that the hybrid model achieves superior robustness and service reliability compared to traditional DRO or learning-only approaches, particularly under high uncertainty. The integration of drones and ground robots further enhances operational flexibility. Further, the results confirm the Hybrid DRO+CB model's suitability for dynamic and uncertain humanitarian logistics scenarios. It delivers both high reliability and efficient performance, with manageable computational cost, making it well-suited for emergency blood delivery and similar high-stakes applications.

Future work can extend this framework in several directions. First, richer context representations and reward functions could improve learning performance. Second, more realistic dynamic simulations, including terrain disruption and vehicle failures, would enhance external validity. Finally, validating the model using real-world operational data would provide stronger empirical grounding and open avenues for deployment.

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